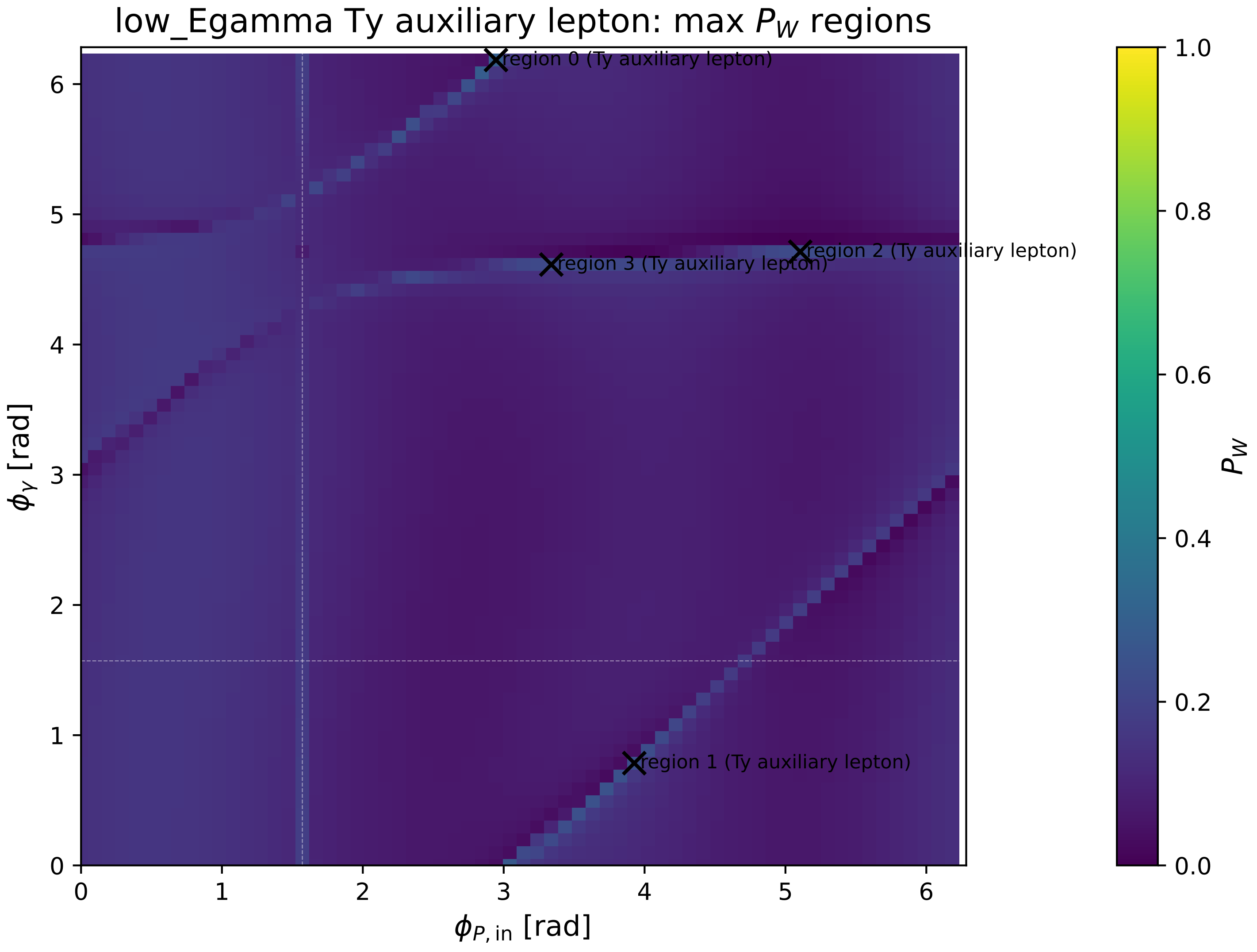
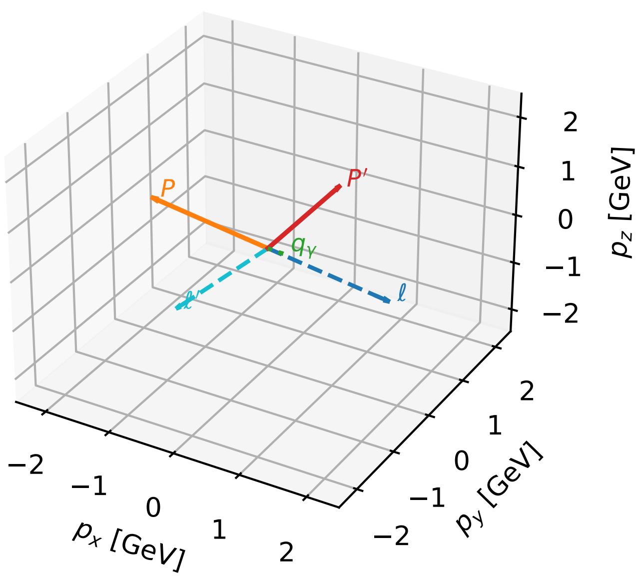


low_Egamma Ty auxiliary lepton: max P_W regions
region 0 (Ty auxiliary lepton)
region 1 (Ty auxiliary lepton)
region 2 (Ty auxiliary lepton)
region 3 (Ty auxiliary lepton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
P_W
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Ty auxiliary lepton characteristic max P_W configuration

3D momenta

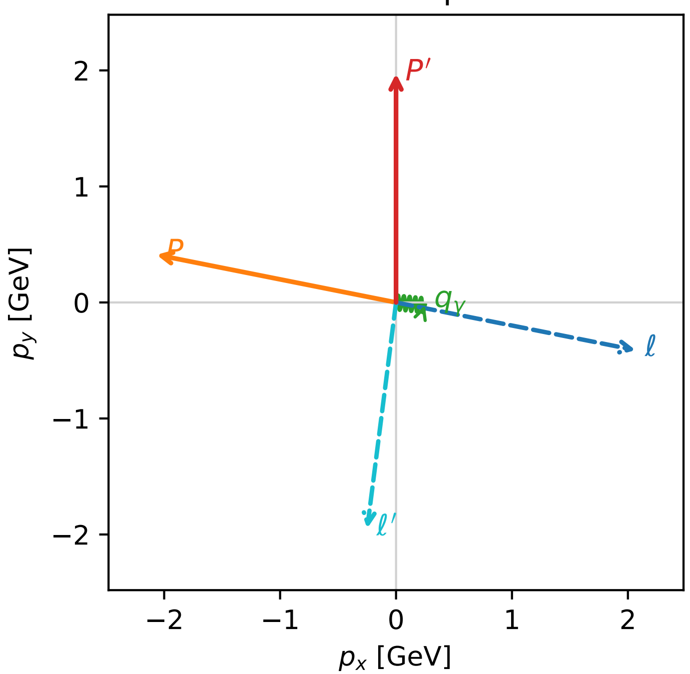


w_purity_low_egamma_ty_electron_region_0 P_W=0.29993
region: low_Egamma_Ty_electron_region_0
outgoing-state purity: 0.529576
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$

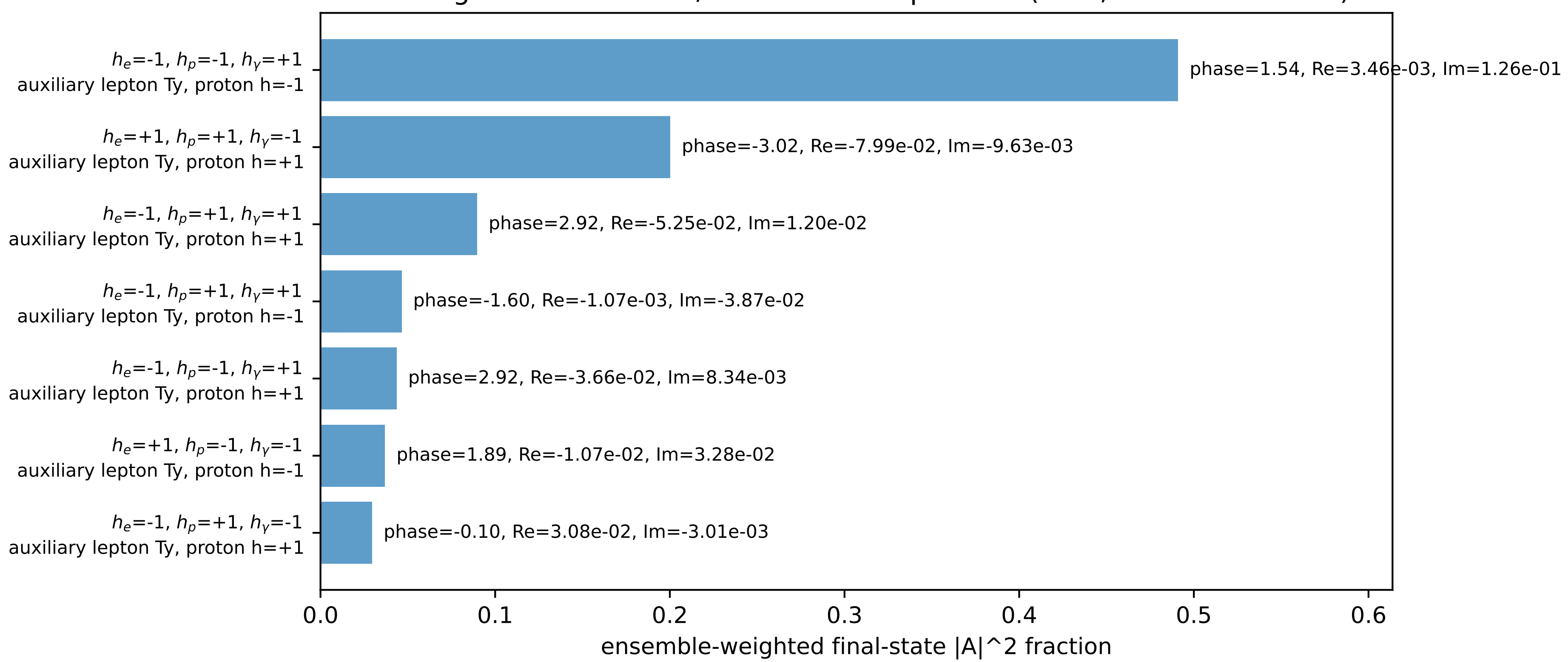
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.97282
θ_{in}=1.57079, ϕ_ℓ=6.08684, ϕ_P=2.94524
E_γ=0.25, ϕ_γ=6.18501
Q^2=7.7072, x_B=0.866355, t=-6.69456, W^2=2.06876, y=0.453051
$\theta(\ell', \gamma)$=91.6522 deg

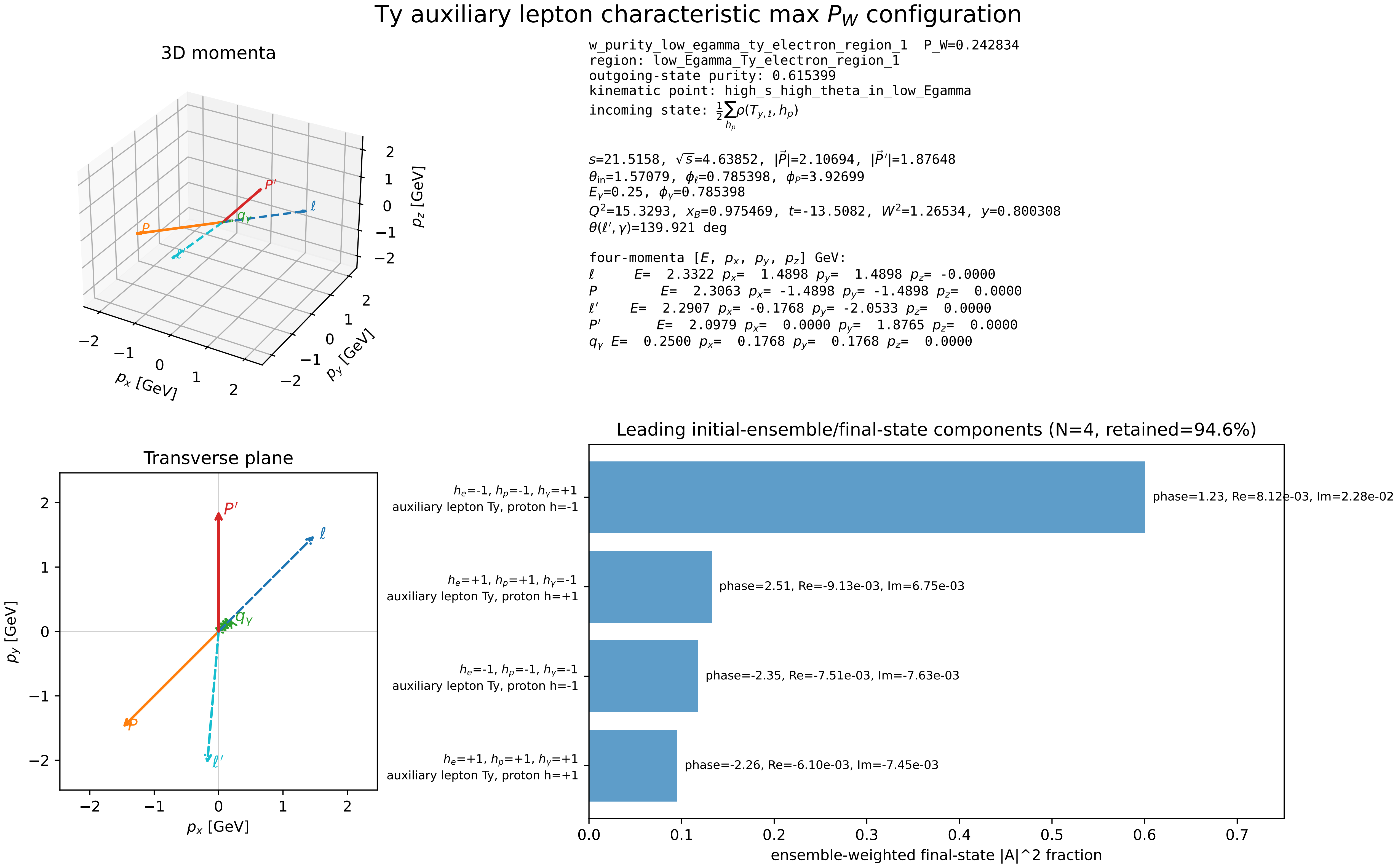
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 2.0665 p_y= -0.4110 p_z= -0.0000
P E= 2.3063 p_x= -2.0665 p_y= 0.4110 p_z= 0.0000
ℓ' E= 2.2041 p_x= -0.2488 p_y= -1.9483 p_z= 0.0000
P' E= 2.1845 p_x= 0.0000 p_y= 1.9728 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2488 p_y= -0.0245 p_z= 0.0000

Transverse plane



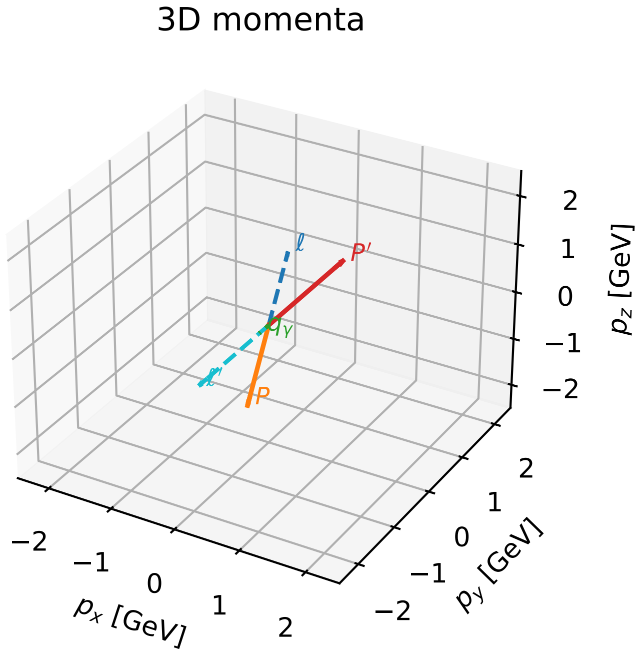
Leading initial-ensemble/final-state components (N=7, retained=93.7%)





Ty auxiliary lepton characteristic max P_W configuration
3D momenta
w_purity_low_egamma_ty_electron_region_1 P_W=0.242834
region: low_Egamma_Ty_electron_region_1
outgoing-state purity: 0.615399
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.87648
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0.785398, ϕ_P=3.92699
E_γ=0.25, ϕ_γ=0.785398
Q^2=15.3293, x_B=0.975469, t=-13.5082, W^2=1.26534, y=0.800308
$\theta(\ell', \gamma)$=139.921 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 1.4898 p_y= 1.4898 p_z= -0.0000
P E= 2.3063 p_x= -1.4898 p_y= -1.4898 p_z= 0.0000
ℓ' E= 2.2907 p_x= -0.1768 p_y= -2.0533 p_z= 0.0000
P' E= 2.0979 p_x= 0.0000 p_y= 1.8765 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1768 p_y= 0.1768 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
ℓ
ℓ'
q_γ
Transverse plane
Leading initial-ensemble/final-state components (N=4, retained=94.6%)
h_e=-1, h_p=-1, h_γ=+1
auxiliary lepton Ty, proton h=-1
phase=1.23, Re=8.12e-03, Im=2.28e-02
h_e=+1, h_p=+1, h_γ=-1
auxiliary lepton Ty, proton h=+1
phase=2.51, Re=-9.13e-03, Im=6.75e-03
h_e=-1, h_p=-1, h_γ=-1
auxiliary lepton Ty, proton h=-1
phase=-2.35, Re=-7.51e-03, Im=-7.63e-03
h_e=+1, h_p=+1, h_γ=+1
auxiliary lepton Ty, proton h=+1
phase=-2.26, Re=-6.10e-03, Im=-7.45e-03
ensemble-weighted final-state |A|^2 fraction

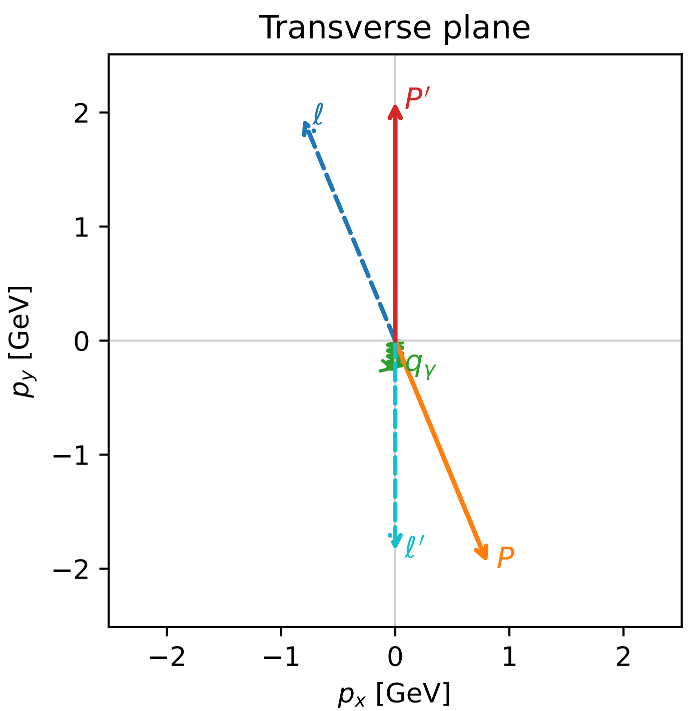
Ty auxiliary lepton characteristic max P_W configuration

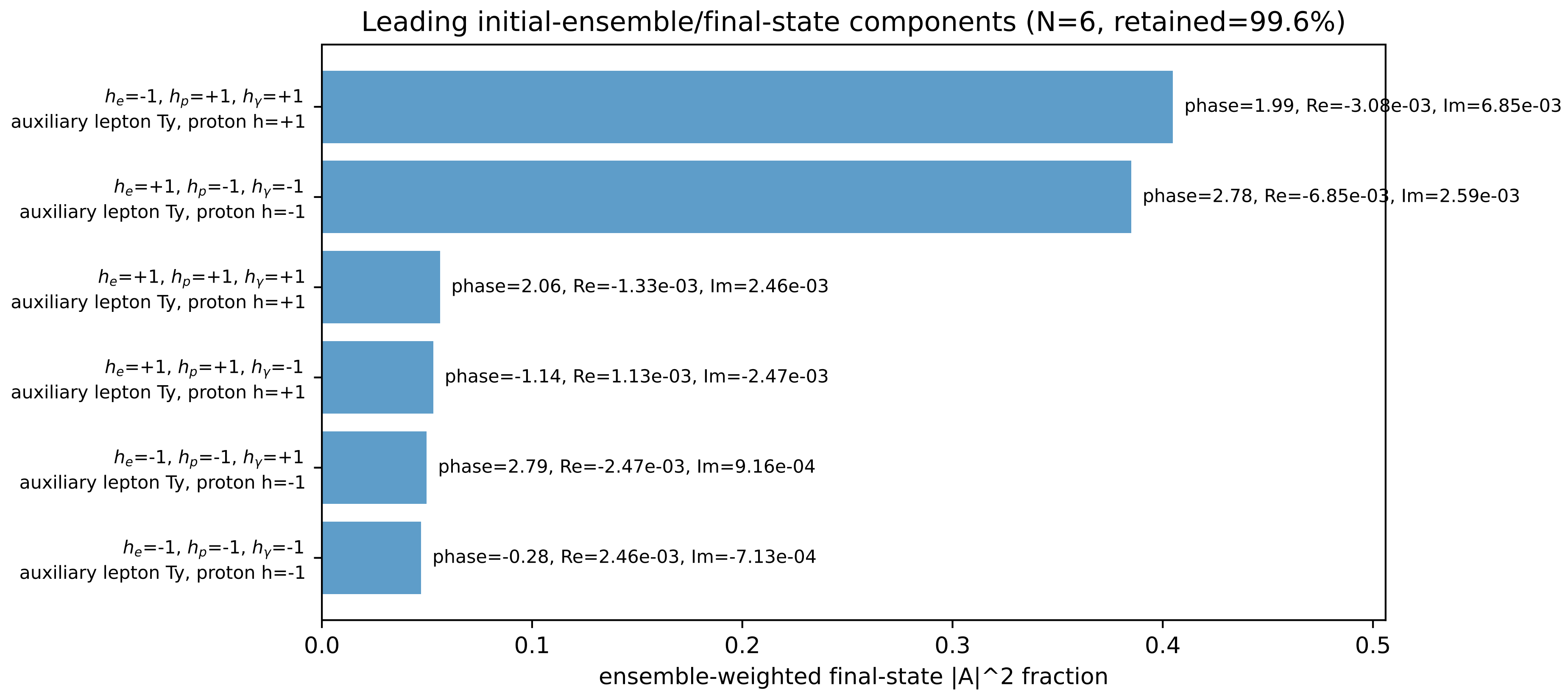


w_purity_low_egamma_ty_electron_region_2 P_W=0.231405
region: low_Egamma_Ty_electron_region_2
outgoing-state purity: 0.500522
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.09195
θ_{in}=1.57079, ϕ_ℓ=1.9635, ϕ_P=5.10509
E_γ=0.25, ϕ_γ=4.71239
Q^2=14.9471, x_B=0.872091, t=-16.9595, W^2=3.07213, y=0.872855
$\theta(\ell', \gamma)$=0 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -0.8063 p_y= 1.9466 p_z= -0.0000
P E= 2.3063 p_x= 0.8063 p_y= -1.9466 p_z= 0.0000
ℓ' E= 2.0959 p_x= 0.0000 p_y= -1.8420 p_z= 0.0000
P' E= 2.2926 p_x= 0.0000 p_y= 2.0920 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.0000 p_y= -0.2500 p_z= 0.0000





Ty auxiliary lepton characteristic max P_W configuration